

LESSON 2.4: Extensibility

What this lesson is about

This lesson introduces customized steps to re-brand Sterling File Gateway user interface, add custom protocols and custom event codes in Sterling File Gateway.

What you should be able to do

- Describe the core Sterling File Gateway application files.
- Explain the procedure to re-brand the user interface.
- Describe the procedure to add custom protocols.
- Explain the procedure to add custom event code.

How you will check your progress

The progress of the course is analyzed based on the successful application of the hands-on exercises.

Additional References

Documentation

- Sterling File Gateway rebrand:
http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg223_updates.doc/SFGRebrandFileGateway.html?lang=en
-

Application file

Introduction

The core Sterling File Gateway application files for file transfers are

- Property file
- Business process
- Database
- Mailbox

Property file

The property files help to customize, and diagnose problems pertaining to the operation of Sterling File Gateway. You can maintain customized property file settings with a customer override property file.

Customer_overrides.properties file overrides the default property settings in the property files. The customer override property file is not changed during the installation of Sterling B2B Integrator upgrades or patches.

Property files are stored in <Install_dir>/properties folder. A few Sterling File Gateway property files are described in the table.

Property File Name	Description
Filegateway_eventcodes.properties	Controls the operation of event codes in Sterling File Gateway.
Filegateway.properties	Controls the operation such as PGP pack and unpack, concurrent session timeout limits during routing, of Sterling File Gateway.
Filegateway_ui.properties	Controls the location of the help systems for Sterling File Gateway, myfilegateway setting.
Filegateway_protocol.properties	Controls the protocols available when specifying the search criteria for FTPIN, FTPOUT, SFTPIN, SFTPOUT.

Application file

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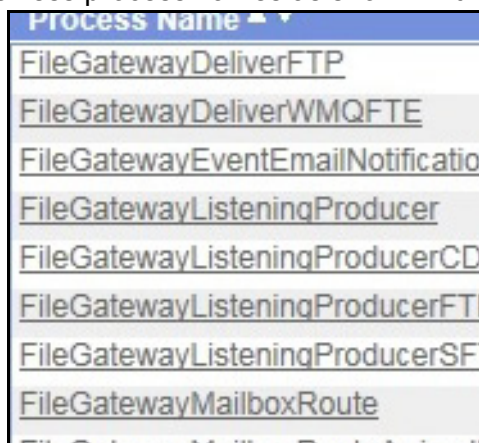
Business process

In Sterling B2B Integrator, a business process is a series of linked software and possibly human activities that accomplishes a business goal. The activities in a business process are primarily performed by components known as services and adapters, and are orchestrated by Sterling B2B Integrator integration engine.

The integration engine runs business processes according to the instructions that comprise your business process models. Technically, a business process model is defined by a BPML file. BPML is an XML-based language used to describe (model) and run business processes.

Complete the followings steps to view the list of system business process of Sterling File Gateway:

1. Login to the filegateway <http://192.168.40.100:9000/filegateway>, by specifying **fg_sysadmin**, as user ID and **password** as password.
2. Navigate to **Tools > B2B Console**.
3. From Administrative console, navigate to **Business Process > Manager**.
4. From the Search panel, type Filegateway and click **GO!**. The system displays a list of Sterling File Gateway business process names as shown in the screen shot..



Process Name
FileGatewayDeliverFTP
FileGatewayDeliverWMQFTE
FileGatewayEventEmailNotification
FileGatewayListeningProducer
FileGatewayListeningProducerCD
FileGatewayListeningProducerFT
FileGatewayListeningProducerSF
FileGatewayMailboxRoute

**Note:**

Do not alter the system business processes. Sterling File Gateway business processes run in queues 3, 4, and 6. Do not change the queue settings of the system business processes because they are configured to optimize queue usage under heavy loads. Custom business processes (used for consumer identification or layer processing) should be configured to execute on any queue except 3 or 4.

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Introduction

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Business process

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The following table provides the description of a few system business processes used for Sterling File Gateway operations:

Business Process	Description
FileGatewayDeliverFTP	Delivers files to consumers using the FTP protocol.
FileGatewayMailboxRoute	Gathers all mailbox messages available for routing.
FileGatewayMailboxRoute ArrivedFile	Routes individual messages identified by FileGatewayMailboxRoute as available for routing.
FileGatewayRoutePGPPa ckageDocument	Routes PGP packaged documents.
FileGatewayRoutePGPU npackageDocument	Routes PGP unpackaged documents.
FileGatewaySendMessage	Sends messages from File Gateway.

Listening Producer API

Listening producer allows Sterling File Gateway to access a user defined server and poll for documents to kick off Sterling File Gateway routes. The current functionality allows for initiating producers, which means you place a file into the producer's mailbox in order to start the transaction.

The list of business processes in Sterling File Gateway 2.2.6.1 for listening producers, mass replay, FTP and SFTP are:

- FileGatewayListeningProducer
- FileGatewayListeningProducerFTP
- FileGatewayListeningProducerSFTP
- FileGatewayProcessFTPReply
- FileGatewayProcessReroute

Introduction

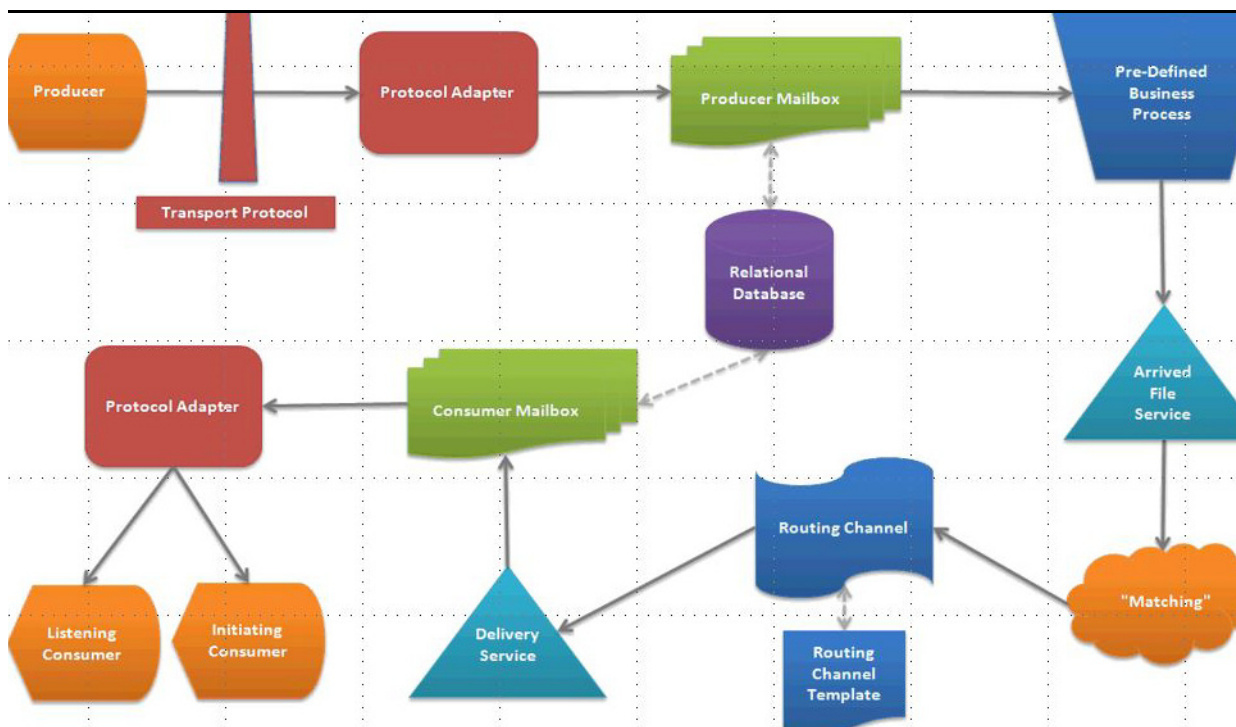
(Continued)

Business process

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Following is a representation of the flow diagram during the file transfer communication:

1. The producer's files are sent through the transport protocol.
2. The transport protocol invokes the appropriate protocol adapter and sends to the producer's mailbox.
3. The system defined business process executes the files from the mailbox.
4. The business process invokes arrived file service, which checks the matching routing channel. The routing channel template verifies and delivers the files.
5. The delivery service, delivers the documents to the consumer mailbox.
6. Consumer mailbox, in-turn, invokes the appropriate protocol adapter and initiates or listens to the document to further integrate into external system or process in Sterling B2B Integrator process.



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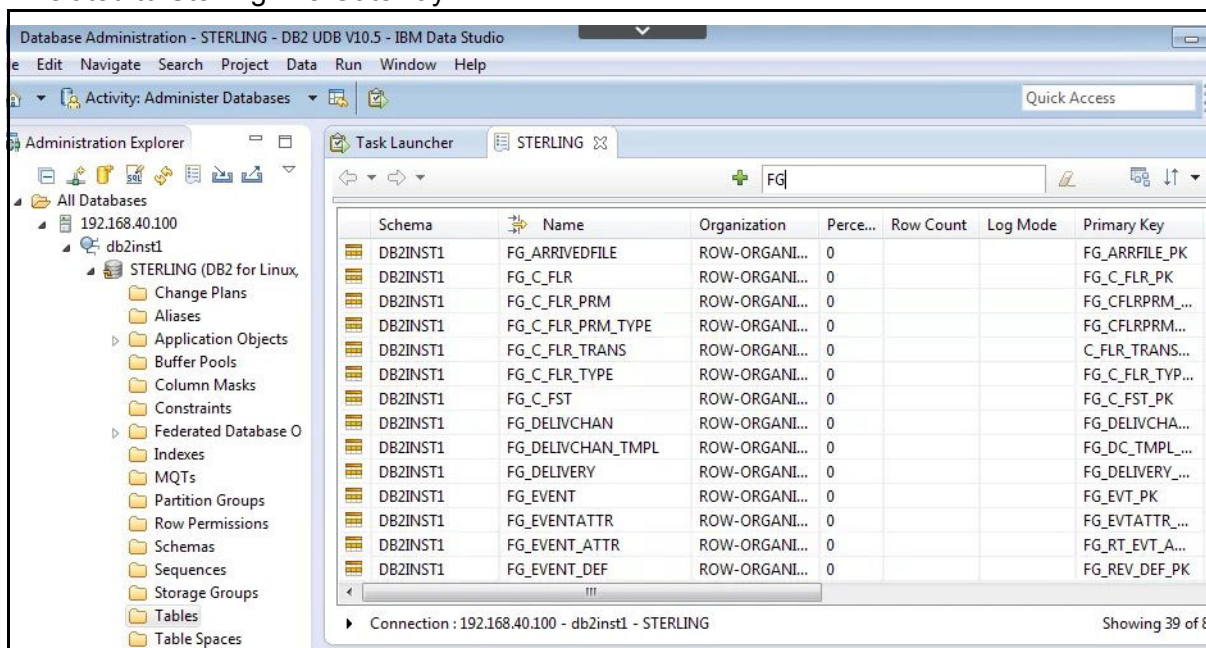
Application file

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Database

Complete the following steps to view the built-in tables of Sterling File Gateway:

1. From the window image, navigate to **Start > All Programs > IBM Data Studio > Data Studio 4.1.0.0 Client**.
2. Expand **All Databases > 192.168.40.100 > Student1 > STERLING DB2 for Linux**.
3. Click **Tables** to view list of tables.
4. Search tables specific to *FG* in the Search icon, as shown in the screen shot to view tables related to Sterling File Gateway.



Note:

There are around 89 built-in tables for Sterling File Gateway application.

Mailbox

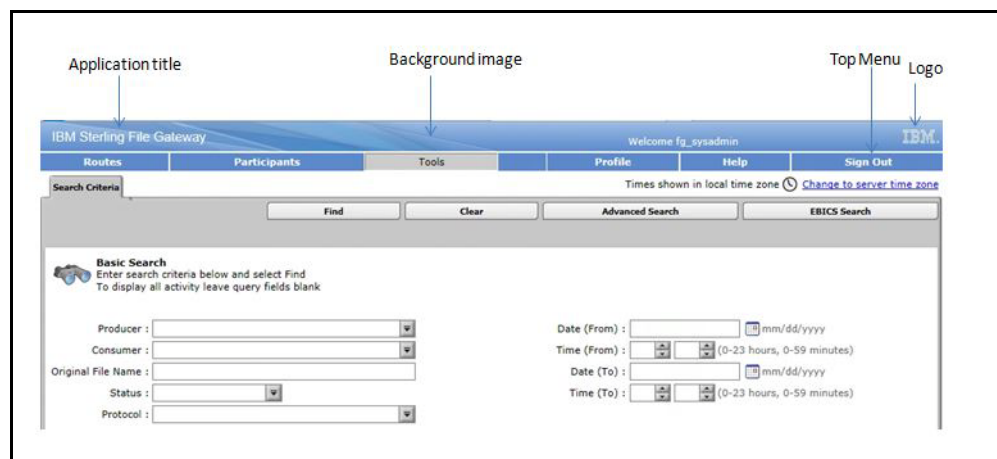
Mailbox concepts are covered in Unit 1 Lesson 4.

Re-branding

Introduction

Sterling File Gateway interfaces can be customized to re-brand the product name, menus of Sterling File Gateway, and myfilegateway user interface to incorporate new company logo or other special content. You can perform re-branding as a process of installation.

The screen shot represents a few standard names such as application title, background image, top menu, logo to customize the user interface of Sterling File Gateway.



Exercise 2.4.1: Re-brand Sterling File Gateway

Introduction

In this exercise, you will customize Sterling File Gateway application to re-brand FileGateway and myfilegateway interfaces.

Instructions

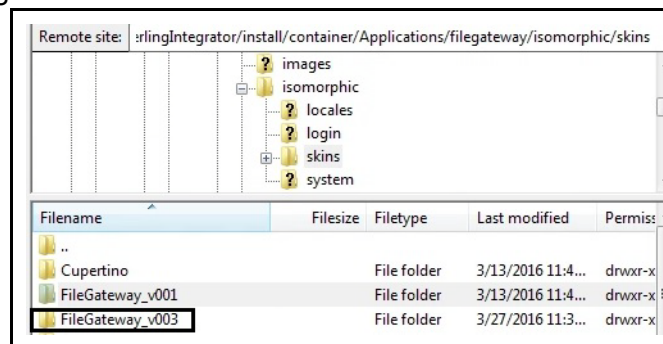
Complete the following steps to rebrand the Sterling File Gateway interface:

Step 1: Access the windows image. From the desktop, open FileZilla.

Step 2: Select `sfg://root@192.168.40.100` from the server drop down list and click Quickconnect. You are now connected to the server image..

Step 1: Navigate to the remote panel to Sterling B2B Integrator properties folder
`/opt/IBM/SterlingIntegrator/install/container/Applications/filegateway/isomorphic/skins/[skin]`

Step 2: Copy the existing FileGateway_v001 skin folder in the `/isomorphic/skins` directory and rename it as FileGateway_vXXX, where XXX is a version number. (You can use any valid directory name.), as highlighted in the screenshot



Step 3: Rename image that is located in `C:/6F86G/ClassFiles` as **hdr_logo.gif** and place it in:

`<install_dir>/container/Applications/<FileGateway_vXXX>/images/FGimages/brandingimages`

An image size of 15 pixels height by 41 pixels width is suggested.

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Exercise 2.4.1: Re-brand Sterling File Gateway

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Instructions

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Step 4: Open the **header.txt** in the **/brandingcontent** directory of the new skin folder that you created. Change the logo reference to:

isomorphic/skins/<FileGateway_vXXX>/images/FGimages/brandingimages/hdr_logo.gif

```

1 <!--
2 Licensed Materials - Property of IBM
3 IBM Sterling File Gateway 2.2
4 (c) Copyright IBM Corp. 2008,2012 All Rights Reserved. US Government Users Restricted Rights - use, duplication
5 -->
6
7 <div id="ibm-banner">
8 <table width="100%" border="0" cellpadding="0" cellspacing="0" style="background: url('isomorphic/skins/FileGateway_v001/images/FGimages/brandingimages/hdr_logo.gif') no-repeat top left;">
9 <tr valign="middle">
10 <td height="35" id="ibm-banner-main" style="width:100%; background: url('isomorphic/skins/FileGateway_v001/images/FGimages/brandingimages/hdr_logo.gif') no-repeat top left;">
11 <span id="ibm-banner-main-product-name" class="ibm-banner-product-name">IBM Sterling File Gateway</span>
12 </td>
13 <td id="ibm-banner-logo" width="49">
14 </td>
15 </tr>
16 </table>
17 </div>

```

Step 5: Open the **customer_overrides.properties** file from **/opt/IBM/SterlingIntegrator/Install/Properties** directory and add the following line and save it.

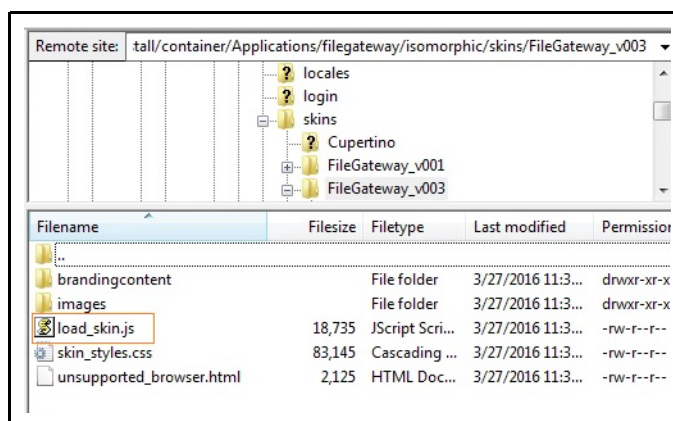
filegateway_ui.fgSkinName=<FileGateway_vXXX>

```

1 mailbox.disallowDuplicateMessages=true
2 filegateway_ui.fgSkinName=FileGateway_v003

```

Step 6: Right click the **load_skin.js** file from **<install_dir>/container/Applications/filegateway/Isomorphic/skins/FileGateway_vXXX** to edit, as shown in the screen shot.



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Implementation

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Exercise 2.4.1: Re-brand Sterling File Gateway

(Continued)

Instructions

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Step 7: Change the **Specify Skin Directory** section to:

```
//-----
// Specify skin directory
//-----
// must be relative to your application file or isomorphicDir
isc.Page.setSkinDir("[ISOMORPHIC]/skins/<FileGateway_vXXX>/")
```

```
// This is used in the branding module
//-----
var myFgHeaderHt = 35;
var myFgHeaderBkClr = "#FFFFFF";
var myFgHeaderALT = "IBM Sterling My File Gateway";
var myFgHeaderUserNameTop = 12;
var myFgHeaderUserNameLeft = "65%";

//-----
// Specify Footer height and color; content is located in "brandingcontent/footer.t
// The ALT is used to render text in case the HTML content is not visible
// This is used in the branding module
//-----
var footerHt = 16;
var footerBkClr = "#4a79c6";
var footerALT = "Copyright IBM Corp. 2008, 2015. All Rights Reserved.";

function loadSkin(theWindow) {
  if (theWindow == null) theWindow = window;
  with (theWindow) {

    //-----
    // Specify skin directory

    // must be relative to your application file or isomorphicDir
    isc.Page.setSkinDir("[ISOMORPHIC]/skins/FileGateway_v003/")
  }
}
```

Exercise 2.4.1: Re-brand Sterling File Gateway

(Continued)

Instructions

....(Continued)

Step 8: Run the following scripts:

```
<install_dir>/bin/hardstop.sh  
<install_dir >/bin/deployer.sh  
<install_dir >/bin/setupfiles.sh  
<install_dir >/bin/run.sh
```

Step 9: Refresh the browser cache.

Step 10: Open Sterling File Gateway. The logo is updated.

Similarly, you can customize the following components of the Sterling File Gateway and myfilegateway. The corresponding component file names are listed against the component:

- Header:
 - [skin]/brandingcontent/header.txt
 - [skin]/brandingcontent/myFgHeader.txt
 - [skin]/load_skin.js
 - [skin]/images/FGimages/brandingimages/...
 - Top menu: [skin]/skin_styles.css
 - Central Panel:
 - [skin]/skin_styles.css
 - [skin]/load_skin.js
 - [skin]/images/...
 - Footer: [skin]/brandingcontent/footer.txt
 - Login Footer: [skin]/brandingcontent/loginFooter.txt
-

Custom Protocols

Introduction

Sterling File Gateway supports to customize protocols for listening consumers.

The information that you provide in customizing protocol procedure determines the text that is shown in the Partner management wizard. Customized protocol, allows you to select, as new choices in the Sterling File Gateway interface:

- when creating a community.
 - when creating a listening consumer in a community.
-

Exercise 2.4.2: Adding custom protocols

Introduction

In this exercise, you will create a custom *Training_HTTP* protocol that to be displayed in the Sterling File Gateway community user interface.

Instructions

Complete the following steps to create custom protocols.

- Step 1:** Check in the **AFTRouteViaHTTP** BPML code from Sterling B2B Integrator business process search panel. The AFTRouteViaHTTP business process allows HTTP protocol replacing RemoteHost, RemotePasswd, RemotePort, and RemoteUserId with valid parameters.
- Step 2:** Use the existing AFTEExtensions.xml as a guide to create an AFTEExtensionsCustomer.xml file to describe the protocol. Do not edit the AFTEExtensions.xml file. The AFTEExtensions.xml file is in the following directory:
`<install_dir>/container/Applications/aft/WEB-INF/classes/resources/xml.`
- Step 3:** Save the *AFTEExtensionsCustomer.xml* file to the same directory as *AFTEExtensions.xml*.
- Step 4:** Uncomment first two elements in the AFTEExtensionsCustomer.xml, that is, filename_suffix and my-protocol-customer.
- Step 5:** Replace the existing bp attribute of my-protocol-customer element with “AFTRouteViaHTTP”.
- Step 6:** Rename **my-protocol-customer** element as “http-customer”.
- Step 7:** Change **mq** group title and labels in AFT extension to “http” labels.
- Step 8:** Use the existing AFTEExtensions.properties file as a guide to create the AFTEExtensionsCustomer.properties file. Do not edit the AFTEExtensions.properties file. The AFTEExtensions.properties is in the following directory:
`<install_dir>\container\Applications\aft\WEB-INF\classes\resources`
- Step 9:** Copy the AFTEExtensionsCustomer.properties file to:
`<install_dir>\container\Applications\aft\WEB-INF\classes\resources`
-

Implementation

Exercise 2.4.2: Adding custom protocols

(Continued)

Instructions

....(Continued)

Step 10: Enter the following data in the new AFTEExtensionsCustomer.properties file.

http.protocol.label.myprotocol.customer = Training_HTTP

http.label.myprotocol.parm1 = HttpSeries ID

http.label.myprotocol.parm2 = HttpSeries name

http.instance.group1.title = Training_HTTP title

Step 11: Ensure that AFTEExtensionsCustomer.xml is error free by opening it in a browser.

Step 12: Run <install_dir>/bin/hardstop.sh.

Step 13: Run <install_dir>/bin/setupfiles.sh.

Step 14: Run <install_dir>/bin/deployer.sh.

Step 15: Run <install_dir>/bin/run.sh.

Step 16: The additional protocol *Training_HTTP* is available when adding and editing communities. After *Training_HTTP* protocol has been chosen for a community, it then becomes available when creating and editing partners in that community.

Verify new HTTP protocol

The additional protocol *Training_HTTP* is available when adding and editing communities. After *Training_HTTP* protocol has been chosen for a community, it then becomes available when creating and editing partners in that community.

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: Navigate to **Participants > Communities**.

Step 3: Click **+add**.

Step 4: Enter **Training_HTTP**.

Step 5: Click **Next**.

Step 6: Select the **Partner listens for protocol connections** check box. The new protocol is listed in the Available protocols list.

Custom event codes

Introduction

Sterling File Gateway allows to add custom events to associate with custom protocols and business processes. You require a code, a name, and a description for each event to create, and permission and subscription settings.

Event Code

The custom event code CUST_0265, consists of keyword (_) underscore followed by a 4-digit code. Four digits are used to identify the attributes for a particular event code.

Keyword - any text, optimally about four characters long, but cannot be FG or AFT.

4-digit Code - consists of two 2-digit codes - a category identifier and a unique identifier. No two events can have the same 4-digit code.

Category Identifier - The first 2- digit code is a category identifier.

Unique Identifier - The second 2-digit code is the unique identifier for the event within the category. 00 - 49 are successful events and 50 - 99 are error events. Error events are highlighted in red in Sterling File Gateway.

- <EventCode>.name - This is the name of your event. **Required.**
- <EventCode>.attributes - A comma-delimited list of attributes without special characters and spaces for the event. The attributes are searchable in the user interface. **Required.**
- <EventCode>.text - This property contains the text that displayed in the user interface when this event is viewed. Generally limited by the Java Format rules. **Required.**
- <EventCode>.description - This description is shown in the subscription UI. **Required.**
- <EventCode>.permissions - This property is a comma-delimited list of permissions for producer, consumer, subscription.

Event Code	Category
00	Producer File Transfer
02	Routing Business Process
04	File Gateway General
05	Route Plan Determination
06	Route Plan Determination
07	Consumer File Transfer

For more details on Sterling File Gateway event codes, refer:

https://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_sys_admin.doc/sfg_abouteventcodes.html?lang=en

Add custom event codes

Introduction

In this exercise you will create a custom event CUST_0265 with code, a name, a description, permission and subscription settings to display the message when the error event from custom business process is generated.

Instructions

Complete the following steps to create custom event code:

Step 1: Open or create a file named `/<install_dir>/properties/customer_overrides.properties`.

Step 2: Add entries in the `customer_overrides.properties` file with the custom event codes. Your new event will be a series of properties describing attributes of the event. The following is a basic template:

```
filegateway_eventcodes.CUST_0265.name=CustomEventFileError
filegateway_eventcodes.CUST_0265.attributes=ConsumerName,
MyServiceResultCode,IDNumber
filegateway_eventcodes.CUST_0265.text=File Error generated during processing:Result
code: {1} reported for File ID {2} received for Consumer {0}
filegateway_eventcodes.CUST_0265.description=Event from Custom BP generated when
error occurs
filegateway_eventcodes.CUST_0265.permissions=producer,consumer, subscription
```



Note: _____

See the `/<install_dir>/properties/filegateway_eventcodes.properties` file as an example for how to structure your event properties. Do not edit the `filegateway_eventcodes.properties` file.

Step 3: Save the `customer_overrides.properties` file and restart Sterling File Gateway for the new file to take effect.

Add custom event codes

(Continued)

Instructions

....(Continued)

Step 4: Create new FileGatewayRouteEventService as shown here.

```
<operation>
  <participant name=" FileGatewayRouteEventService "/>
    <output message="Output">
      <assign to="EventCode">CUST_0265</assign>
      <assign to="ExceptionLevel">Normal</assign>
      <assign to="EventAttributes/Directory" from="directory/text() " append="true"/>
      <assign to="EventAttributes/Comment" > BP changed directories</assign>
      <assign to="." from="*"></assign>
    </output>
    <input message="Input">
      <assign to="." from="*"></assign>
    </input>
  </operation>
```

Step 5: Edit your business process that generates the new event codes to invoke the FileGatewayRouteEventService, with appropriate parameters, including the new custom event codes.

Step 6: Check in your business process that contains the FileGatewayRouteEventService, with appropriate parameters and including the new custom event code. Sterling File Gateway must start the business process through its extension capability for consumer identification or custom protocol.

Step 7: Log in to myfilegateway and ensure that you have received the CUST_0265 notification.

Lesson review

What you have been able to do

- Describe the core Sterling File Gateway application files.
 - Explain the procedure to re-brand the user interface.
 - Describe the procedure to add custom protocols.
 - Explain the procedure to add custom event code.
-

Course wrap-up

Course wrap-up enables you to submit an evaluation of the class and provides information on additional training options.

Introduction

This course described about the terms, concepts, functionality, installation, user interfaces, configuring users and trading partners of Sterling File Gateway. You have learned to create and use routing channel templates, manage mailboxes and create static and dynamic routing channel template, processing various file layers and generate reports. You have also learned extensibility of Sterling File Gateway application.

Completed objectives

Completed course objectives

This course was designed to enable you to:

- Install and configure Sterling File Gateway.
 - Describe community, groups and users.
 - Build static and dynamic routing channel templates.
 - Describe the different persona's and their responsibilities.
 - Configure routing channels.
 - Transfer file from a producer to a consumer.
 - Broadcast messages.
 - Explain the enhanced visibility to track a route.
 - Explain container and non-container layers.
 - Generate reports.
 - Discuss the procedure for Sterling File Gateway extensibility.
 - Discuss the procedure to rebrand Sterling File Gateway.
-

Course evaluation

Completing the evaluation

To ensure that the IBM Managing Sterling File Gateway V2.6.6.1 is as effective as possible and meets the needs of customers, please share your feedback. Feedback on the site, curriculum, and instructor tell us what was good about the class and what can be improved.

Take the time to complete the online course evaluation on the IBM Training website:

https://survey6.spss-asp.com/mriweb/mriweb.dll?i.project=ibmlearningsrvs_learningsservicebasesurveyjuly2013

Your instructor provides you with the class number and course code to complete the access the online course evaluation.

Thank you for your valuable inputs.

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Thank you for your valuable inputs.

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